

Topology

Home Work 1

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Exercise 1 - Attempted

Verify that each of the following functions defines a metric.

a) Let X be any set and define $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ by

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

b) Let $C([0, 1], \mathbb{R})$ be the space of continuous real-valued functions on $[0, 1]$. Show that the L^1 -distance

$$d_1(f, g) = \|f - g\|_1 = \int_0^1 |f(t) - g(t)| dt$$

defines a metric on $C([0, 1], \mathbb{R})$.

ANSWER:

For any metric space (X, d) we know that the metric must satisfy the following 3 distance relations:

1. $d(x, y) \geq 0 \quad \forall x, y \in X$ and also: $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x) \quad \forall x, y \in X$
3. $d(x, y) \leq d(x, z) + d(z, y) \quad \forall x, y, z \in X$

(a) For the Trivial Metric we must show that each of the above 3 conditions hold.

(i) $d(x, y) \geq 0$ must hold as the only possible values are 0 and 1.

$x = y \iff d(x, y) = 0$ follows directly from the definition.

(ii) If $x = y \implies d(x, y) = d(y, x) = 0$. If $x \neq y \implies d(x, y) = 1$ and $d(y, x) = 1$.

As these are the only possible case we have that $d(x, y) = d(y, x)$.

(iii) We have two cases to consider:

(A) $d(x, y) = 0 \implies x = y$. In this case $d(x, z) + d(z, y)$ can't be less than zero. Hence, inequality (3) must hold true.

(B) $d(x, y) = 1 \implies x \neq y$. In this case $d(x, z) + d(z, y)$ can't be less than 1.

If $x = z$, we can't have $z = y$ since $x \neq y$. The reverse is also true. Hence we must have $d(x, z) + d(z, y) \geq 1$, hence inequality (3) must hold true.

As all 3 conditions hold (X, d) is a valid metric space.

(b) We again check each of the above conditions (1)-(3) for a metric.

(i)

$$\|f - g\|_1 = 0 \implies f(x) = g(x) \quad \forall x \in [0, 1]$$

$f, g \in C[0, 1] \implies (f - g) \in C[0, 1]$. Assume $f(\bar{u}) \neq g(\bar{u})$ at some $\bar{u} \in [0, 1]$, then $f(x) \neq g(x)$ in some interval $(\bar{u} - \delta, \bar{u} + \delta)$ ($\delta > 0$).

Hence:

$$|f(x) - g(x)| > 0 \quad \text{for } x \in (\bar{u} - \delta, \bar{u} + \delta).$$

But then

$$\|f - g\|_1 = \int_0^1 |f(x) - g(x)| dx > 0$$

contradiction. Hence:

$$f(x) = g(x), \quad \forall x \in [0, 1].$$

On the other hand, if we now assume that:

$$f(u) - g(u) = 0, \quad \forall u \in [0, 1],$$

then

$$\int_0^1 |f(u) - g(u)| du = \int_0^1 0 du = 0$$

and hence

$$\|f - g\|_1 = 0.$$

(ii)

$$\begin{aligned} \|f - g\|_1 &= \int_0^1 |f(u) - g(u)| du \\ &= \int_0^1 |g(u) - f(u)| du \\ &= \|g - f\|_1. \end{aligned}$$

(iii) For all $u \in [0, 1]$ and $h \in C[0, 1]$:

$$0 \leq |f(u) - g(u)| \leq |f(u) - h(u)| + |h(u) - g(u)|.$$

Hence, integrating both sides:

$$\int_0^1 |f(u) - g(u)| \, du \leq \int_0^1 |f(u) - h(u)| \, du + \int_0^1 |h(u) - g(u)| \, du$$

and therefore

$$\|f - g\|_1 \leq \|f - h\|_1 + \|h - g\|_1.$$

We again see that we have a valid distance function and hence a valid metric space.

Exercise 2 - Not Attempted

Which of the following functions $d : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$ are valid metrics on $\mathbb{R}^2$?

For those that are, what does the unit open ball $B_d(x, 1)$ look like?

Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be two vectors in $\mathbb{R}^2$.

i)

$$d(x, y) = |x_1 - y_1| + |x_2 - y_2|.$$

ii)

$$d(x, y) = \min \{|x_1 - y_1|, |x_2 - y_2|\}.$$

iii)

$$d(x, y) = (x_1 - y_1)^2 + (x_2 - y_2)^2.$$

iv) $d(x, x) = 0$, and if $x \neq y$, then

$$d(x, y) = \|x\|_2 + \|y\|_2.$$

(Recall that on $\mathbb{R}^2$,

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2}.$$

)

Exercise 3 - Attempted

Let (X, d) be a metric space. Recall that a set V is closed if and only if its complement,

$$V^c = X \setminus V,$$

is open in (X, d) .

- i) Show that a finite union of closed sets is closed.
- ii) Show that an arbitrary intersection of closed sets is closed.
- iii) Give an example where a countable union of closed sets is not closed.
- iv) Check that closed balls in (X, d) are closed.

ANSWER:

(i) Let $(V_i)_{i=1}^n$ be a finite collection of closed sets. Let $U_i = X \setminus V_i$ be the corresponding open sets. We know that the finite intersection of open sets is again open. I.e.:

$$J = \bigcap_{i=1}^n U_i$$

is open. But this means that J^c must be closed and hence that:

$$\begin{aligned} J^c &= \left[\bigcap_{i=1}^n U_i \right]^c \\ &= \bigcup_{i=1}^n U_i^c \quad (\text{De Morgan's Law}) \\ &= \bigcup_{i=1}^n V_i \end{aligned}$$

is closed.

We conclude that the union of a finite number of open sets is again open.

(ii) Let $(V_i)_{i \in I}$ be an arbitrary collection of closed sets - indexed by I . Again, let $U_i = X \setminus V_i$ be the corresponding open sets. Now we know that any union of open sets is open. Hence:

$$J = \bigcup_{i \in I} U_i$$

is open. Consequently, its complement must be closed and hence:

$$\begin{aligned}
J^c &= \left[\bigcup_{i \in I} U_i \right]^c \\
&= \bigcap_{i \in I} U_i^c && \text{(De Morgan's Law)} \\
&= \bigcap_{i \in I} V_i
\end{aligned}$$

is closed.

So, the final result shows that the intersection of any collection of closed sets is again closed.

(iii) Consider the collection of open sets $(U_i)_{i \in \mathbb{N}}$, where: $U_n = (2 - \frac{1}{n}, 4 + \frac{1}{n})$. The intersection of all such open sets is the closed set $[2, 4]$.

(iv) To show that $\hat{B}(x, r) := \{y \in X \mid d(x, y) \leq r \ (r > 0)\}$ is closed we just need to demonstrate that its complement is open. Define $J = X \setminus \hat{B}(x, r)$ as this complement. J is open if, given any $z \in J$, we can determine an $\epsilon > 0$ such that $B(z, \epsilon) \subset J$.

Now, given any $z \in J$, it must be the case that $d(x, z) = s > r$. Let $\epsilon = (s - r) > 0$. Next, take any $p \in B(z, \epsilon)$. Then, by the triangle inequality:

$$\begin{aligned}
d(x, z) &\leq d(x, p) + d(p, z) \\
&\leq d(x, p) + d(z, p) \\
&< d(x, p) + \epsilon && \text{(since } d(p, z) < \epsilon) \\
\implies d(x, z) - \epsilon &< d(x, p) \\
\implies s - (s - r) &< d(x, p) && \text{(As } \epsilon = s - r) \\
\implies r &< d(x, p) \\
\implies p &\notin \hat{B}(x, r) \\
\implies p &\in J
\end{aligned}$$

As this is true $\forall p \in B(z, \epsilon)$ we must conclude that: $B(z, \epsilon) \subset J$, and J is open. □

Exercise 4 - Attempted

Determine whether the following subsets of $\mathbb{R}^2$ are open or closed.

i) $D = \{\text{the } x\text{-axis}\}$.

ii)

$$J = (0, 1) \times \{0\} = \{(x, 0) : 0 < x < 1\}.$$

iii)

$$F = \{(x, y) : x \text{ and } y \text{ are integers}\}.$$

iv)

$$G = \left\{ (1, 0), \left(\frac{1}{2}, 0\right), \left(\frac{1}{3}, 0\right), \dots, \left(\frac{1}{n}, 0\right), \dots \right\}.$$

v) $H = \mathbb{R}^2$.

ANSWER:

Here I give a "rough" argument rather than the details - which directly follow from the arguments.

(i) The set D is closed because its complement $D^c = X \setminus D$ consists of the two open half planes in $\mathbb{R}^2$: $y > 0$ and $y < 0$ - and the union of two open sets is open hence $D^c = X \setminus D$ is open. **Thus, D must be closed.**

(ii) The set J is neither open nor closed. As J is effectively 1-dimensional it is not possible for it to contain any ball in $\mathbb{R}^2$ - as this would be 2-dimensional - so it can't be open. Also J^c contains the point $(1, 0)$. There is no open ball at this point that doesn't intersect J . Hence, J^c can't be open. So, J can't be closed.

(iii) The set F is closed. Consider the complement set $J = \mathbb{R}^2 \setminus F$ and any $p \in J$. Then there is a positive minimum distance, s , from any p to the points in F . Clearly $B(p, s) \subset J$. Hence J is open and hence F is closed.

(iv) The set G is neither open nor closed. Again consider $J = \mathbb{R}^2 \setminus G$. Then it is clear that $0 \in J$, but no open ball centred at 0 is completely contained in J - so J is not open and hence G is not closed. Clearly there is no open ball about any point of G that is contained in G - so G is also not open.

(v) The set H is open. Take any point $p \in H$, then $B(p, 1) \subset H$ and hence H is open.

Exercise 5 - Attempted

- i) Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$. Prove that f is continuous in the topological sense, i.e., the preimage of every open set is open, if and only if it satisfies the classical epsilon-delta definition of continuity:

$$\forall x \in X \ \forall \varepsilon > 0 \ \exists \delta > 0 \text{ s.t. } \forall x' \in B_{d_X}(x, \delta), \ d_Y(f(x), f(x')) < \varepsilon.$$

- ii) Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$. Prove that f is continuous if and only if the preimage of every closed set is closed.

ANSWER:

We are given a function $f : X \rightarrow Y$; where (X, d_X) and (Y, d_Y) are both metric spaces.

(i) First we assume the $\epsilon - \delta$ definition of continuity - which we proved in class.

Let $U \subset Y$ be open in (Y, d_Y) . Let $V = f^{-1}(U)$. If $V = \emptyset$, then this is OK as the empty set is open.

If $V \neq \emptyset$, take any $x \in V$ with $f(x) \in U$. As U is open there exists an $\epsilon > 0$ such that: $B_{d_Y}(f(x), \epsilon) \subset U$.

Using the $\epsilon - \delta$ definition of continuity we can find a $\delta > 0$ such that:

$$f(B_{d_X}(x, \delta)) \subset B_{d_Y}(f(x), \epsilon) \implies B_{d_X}(x, \delta) \subset f^{-1}(U) = V$$

That is V is open.

Next we assume the "preimage of every open set is open" definition of continuity - which we didn't prove in class.

Let $U \subset Y$ be open in Y . Then $f^{-1}(U) = V \subset X$ is open by assumption. If $V = \emptyset$ there is nothing to prove (as empty set is open).

So, assume that $V \neq \emptyset$. Take any $x \in V \implies f(x) \in U$. As U is open: $\exists \epsilon > 0$ such that $B_{d_Y}(f(x), \epsilon) \subset U$.

But we also have that $f^{-1}(B_{d_Y}(f(x), \epsilon)) = T \subset V$, where $x \in T$ and T will be open, by assumption. Hence $\exists \delta > 0$ such that: $B_{d_X}(x, \delta) \subset T$. Then $f(B_{d_X}(x, \delta)) \subset f(T) \subset B_{d_Y}(f(x), \epsilon)$.

This is the " $\epsilon - \delta$ " definition of continuity.

(ii) We assume the "preimage of every open set is open" definition of continuity.

Let $D \subset Y$ be closed. Then $D^c \subset Y$ is open. Consequently: $f^{-1}(D^c) \subset X$ is also open by assumption. Then:

$$\begin{aligned} f^{-1}(D^c) &= f^{-1}(Y \setminus D) \\ &= f^{-1}(Y) \setminus f^{-1}(D) \text{ (using the identity: } f^{-1}(A \setminus B) = f^{-1}(A) \setminus f^{-1}(B)) \\ &= X \setminus f^{-1}(D) \end{aligned} \tag{5.1}$$

We know that the set shown in (5.1) is open. This means that $f^{-1}(D)$ is closed. That is, the preimage of every closed set is closed".

□

Exercise 6 - Not Attempted

Which letters and digits are homeomorphic to each other, written as below?

A B C D E F G H J K L M N O P Q R
S T U V W X Y Z 0 1 2 3 4 5 6 7 8 9

You don't need to prove your answer rigorously—you don't have the tools for that yet; this exercise is just to build up an intuition for homeomorphism informally!